

PD-(L)1: 13 molecules stopped early

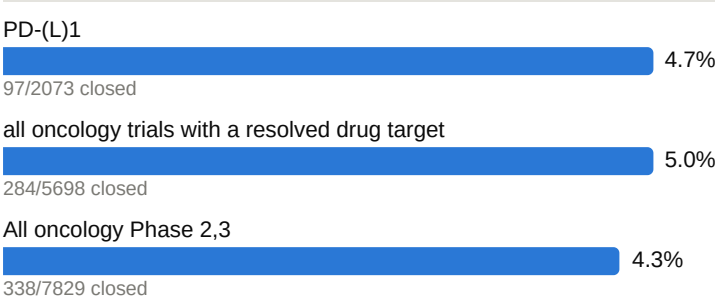
97 stopped trials across 13 molecules: Pembrolizumab, Nivolumab, Atezolizumab, Avelumab and 9 more. 39 of the stops were the trial's own verdict and 12 followed a decision taken elsewhere; 46 cannot be established. Against 5.0% for all oncology trials with a resolved drug target and 4.3% across all oncology Phase 2,3 trials in the same window. The rate counts trials that stopped early for efficacy, safety or benefit–risk reasons. Trials that ran to completion and missed their primary endpoint are counted separately below, never in the rate.

4.7% 97 of 2073 closed trials (95% CI 3.9%–5.7%)	75 / 18 / 1 efficacy / safety / both (adds to 97)	36.5% of 5682 trials have closed · 3609 still open or unresolved	1.5% stopped for a biological reason within 3 years of starting (95% CI 1.1%–1.9%; 2671 still at risk; 1.5% area-wide)
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63 of 147

Completed and missed the primary endpoint. Of 1469 completed trials in this segment, 147 posted a primary superiority analysis on ClinicalTrials.gov; 63 came back non-significant and 59 significant, 25 split across co-primary endpoints. Counted separately and never in the rate above. The rest posted no analysis to read, so this is a floor, not a rate.

How this compares



What is in this class: 1087 distinct experimental drugs across 5682 trials (antibody 5387, small molecule 2617, unknown 974, protein 321). Most tested: PEMBROLIZUMAB (1321), NIVOLUMAB (911), DURVALUMAB (463), ATEZOLIZUMAB (434), PACLITAXEL (416). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 97 stops came from 89 sponsor-asset programmes across 65 sponsors; the largest (Roche) contributed 4; removing that programme's trials from both sides leaves 4.6%. Screened alongside 85 other segments in this area, it does not survive a 10% false-discovery correction (q=1, Benjamini-Yekutieli). A further 122 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 10.6%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2015–2017	29	732	4.0%	2.8%–5.6%
2018–2020	44	865	5.1%	3.8%–6.8%
2021–2024	24	476	5.0%	3.4%–7.4%

Where the stops sit

Efficacy / futility: 75, Safety: 18, Benefit–risk: 3, Efficacy and safety: 1.
Sponsors with most stops: Pfizer (7), Roche (6), Novartis (4), National Cancer Institute (NCI) (4), Merck KGaA (3), Aduro Biotech, Inc. (3).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT05860296	2	SignalChem Lifesciences Corporation	SLC-391, PEMBROLIZUMAB	2025-12	Efficacy	Lack of efficacy and enrollment challenges

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT03274258	2	M.D. Anderson Cancer Center	IPILIMUMAB, NIVOLUMAB	2025-11	Efficacy	The study met its prespecified futility criteria during the prespecified interim futility analysis after enrolling the first cohort of 10 patients.
NCT03225547	2	University of Chicago	PEMBROLIZUMAB, MIFEPRISTONE	2025-09	Safety	side effects (rash)
NCT05491317	2	Genmab GMAB	Tecaginlimab, PEMBROLIZUMAB	2025-08	Efficacy	After reviewing all available data from Part 1 of the trial, the Sponsor has decided not to proceed to Part 2 (randomized Phase 2) of the trial due to lack of efficacy
NCT05617313	2	Jeffrey Clarke	PEMBROLIZUMAB, Anti-CFH Monoclonal Antibody GT10...	2025-06	Efficacy	Study did not meet the criteria for continuation after interim analysis
NCT04955743	2	Yale University	PEMBROLIZUMAB, LENVATINIB	2025-05	Efficacy	Lack of activity in the drugs

The 6 most recent of 97 stopped trials. Every molecule behind all 97 is named above; the complete trial list, the trials still running and the terminations with no readable cause ship with the dataset and the evidence package.

Method

Denominator: ClinicalTrials.gov interventional Phase 2,3 oncology trials started 2015–2024 that have closed (completed or terminated).
Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 76% of industry oncology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits

Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary analysis that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 36.5% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

The whole cohort, trial by trial. The evidence package names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. €99, delivered immediately: clinicaltrialfailures.com/asset-check

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.